

Why Node2Vec Resists Theoretical Analysis: Tracing Community Information Through Graph Embedding Pipelines

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1 Introduction

Spectral methods like Laplacian Eigenmaps have well-understood theoretical properties on generative models like the Degree-Corrected Stochastic Block Model (DCSBM). In particular, Qin and Rohe [2013] showed that eigenvector normalization perfectly separates community from degree information—community lives in direction, degree in magnitude.

Neural embedding methods like Node2Vec lack comparable theoretical guarantees. This paper asks: **why?** We trace community information through the Node2Vec pipeline and identify precisely where theoretical analysis breaks down.

Main contributions:

1. We show the random walk co-occurrence matrix preserves the spectral structure of the normalized adjacency via a similarity transformation (Section 4).
2. We prove the PMI transform converts the multiplicative community-degree structure to additive, breaking the separation property (Section 5).
3. We identify three theoretical barriers in SGNS that prevent closed-form characterization of embeddings (Section 6).

2 Background: DCSBM and Spectral Methods

2.1 The Degree-Corrected Stochastic Block Model

In the DCSBM, each node i has community assignment $g_i \in \{1, \dots, K\}$ and degree parameter $\theta_i > 0$. Edges form independently with $\mathbb{E}[A_{ij}] = \theta_i \theta_j B_{g_i, g_j}$, where B is the community connectivity matrix.

The expected degree factors as $\mathbb{E}[d_i] = \theta_i \cdot \gamma_{g_i}$, where $\gamma_g = \sum_k B_{gk} \text{Vol}_k$ and $\text{Vol}_k = \sum_{j \in \text{Comm}_k} \theta_j$.

2.2 Known Result: Spectral Separation

For the normalized adjacency $S = D^{-1/2} A D^{-1/2}$, Qin and Rohe [2013] established:

Theorem 1 (Qin and Rohe 2013). *In the DCSBM population limit, eigenvectors of S have the form $u_i = \sqrt{\theta_i} \cdot c_{g_i}$, where c_g depends only on community. Consequently, normalized embeddings $\hat{u}_i = u_i / \|u_i\|$ depend only on community g_i .*

This clean separation—community in direction, degree in magnitude—enables perfect community recovery via row-normalized spectral clustering.

3 The Node2Vec Pipeline

Node2Vec proceeds through four stages:

$$\text{Graph } G \xrightarrow{\text{random walks}} C \xrightarrow{\text{PMI}} M \xrightarrow{\text{SGNS}} E \quad (1)$$

We trace community information through each stage.

4 Stage 1-2: Co-occurrence Preserves Structure

Random walks on G generate node sequences. The co-occurrence matrix C counts pairs appearing within a context window of size w . Following Qiu et al. [2018]:

$$C = \sum_{r=1}^w T^r D^{-1} \quad (2)$$

where $T = D^{-1}A$ is the transition matrix.

Proposition 1 (Co-occurrence preserves eigenvector structure). *The matrix $\tilde{C} = D^{1/2}CD^{1/2}$ satisfies $\tilde{C} = \sum_{r=1}^w S^r$, a polynomial in S .*

Proof. We have $D^{1/2}TD^{-1/2} = D^{-1/2}AD^{-1/2} = S$. Therefore:

$$D^{1/2}CD^{1/2} = \sum_{r=1}^w D^{1/2}T^r D^{-1}D^{1/2} = \sum_{r=1}^w D^{1/2}T^r D^{-1/2} = \sum_{r=1}^w S^r \quad (3)$$

□

Implication: Since $\tilde{C} = p(S)$ for polynomial p , they share eigenvectors. The matrix C is similar to $\tilde{C}$, so its eigenvectors are $D^{-1/2}\mathbf{u}$, which still factor as:

$$[D^{-1/2}\mathbf{u}]_i = \frac{\sqrt{\theta_i} \cdot c_{g_i}}{\sqrt{d_i}} = \frac{\sqrt{\theta_i} \cdot c_{g_i}}{\sqrt{\theta_i \gamma_{g_i}}} = \frac{c_{g_i}}{\sqrt{\gamma_{g_i}}} \quad (4)$$

This depends only on community—the multiplicative structure is preserved.

5 Stage 3: PMI Transform Breaks Separation

Skip-Gram with Negative Sampling (SGNS) implicitly factorizes the PMI matrix [Levy and Goldberg, 2014]:

$$M_{ij} = \log(C_{ij}) - \log(\pi_i) - \log(\pi_j) + \text{const} \quad (5)$$

where $\pi_i \propto d_i$ is the stationary distribution.

5.1 The Logarithm Converts Multiplicative to Additive

From Section 4, the co-occurrence matrix has entries:

$$C_{ij} = \frac{[p(S)]_{ij}}{\sqrt{d_i d_j}} \quad (6)$$

The matrix $p(S)$ inherits spectral structure from S . Using Theorem 1, the leading terms satisfy:

$$[p(S)]_{ij} \approx \sqrt{\theta_i \theta_j} \cdot F_{g_i, g_j} \quad (7)$$

where F_{g_i, g_j} depends only on communities.

Therefore:

$$C_{ij} \approx \frac{\sqrt{\theta_i \theta_j} \cdot F_{g_i, g_j}}{\sqrt{\theta_i \gamma_{g_i}} \sqrt{\theta_j \gamma_{g_j}}} = \frac{F_{g_i, g_j}}{\sqrt{\gamma_{g_i} \gamma_{g_j}}} \quad (8)$$

Key observation: The θ_i terms cancel in C_{ij} —the co-occurrence matrix depends only on community structure, not individual degree parameters.

5.2 PMI Reintroduces Degree Dependence

The PMI transform subtracts $\log(\pi_i) + \log(\pi_j)$ where $\pi_i \propto d_i = \theta_i \gamma_{g_i}$:

$$M_{ij} = \log(C_{ij}) - \log(d_i) - \log(d_j) + \text{const} \quad (9)$$

$$= \log(C_{ij}) - \log(\theta_i) - \log(\gamma_{g_i}) - \log(\theta_j) - \log(\gamma_{g_j}) + \text{const} \quad (10)$$

Since C_{ij} depends only on communities, we have:

$$M_{ij} = \underbrace{H_{g_i, g_j}}_{\text{community block}} - \underbrace{\log(\theta_i) - \log(\theta_j)}_{\text{degree offsets}} + \text{const} \quad (11)$$

where $H_{g_i, g_j} = \log(C_{ij}) - \log(\gamma_{g_i}) - \log(\gamma_{g_j})$ depends only on communities.

5.3 Why This Breaks Separation

	Structure	Normalization separates?
Laplacian eigenvectors	$u_i = \sqrt{\theta_i} \cdot c_{g_i}$ (multiplicative)	Yes
PMI matrix	$M_{ij} = H_{g_i, g_j} - \log \theta_i - \log \theta_j$ (additive)	No

In the Laplacian, dividing by magnitude removes $\sqrt{\theta_i}$, leaving only c_{g_i} .

In PMI, the degree terms $-\log(\theta_i)$ are *additive offsets* to each row/column. No normalization of the resulting embeddings can remove additive structure—it would require *subtraction*, not division.

5.4 Rank Structure of PMI

The PMI matrix decomposes as:

$$M = H^{(g)} + \mathbf{r} \mathbf{1}^T + \mathbf{1} \mathbf{r}^T + \text{const} \quad (12)$$

where $H^{(g)}$ is rank- K (community block structure) and $r_i = -\log(\theta_i)$ creates rank-2 offset terms.

The total rank is at most $K + 2$, but the community and degree information are now additively entangled rather than multiplicatively separable.

6 Stage 4: SGNS Optimization

Even with the PMI matrix fully characterized, three barriers prevent determining where community information ends up in learned embeddings.

6.1 Barrier 1: Non-uniqueness of Factorization

SGNS seeks embeddings E, E' such that $E_i \cdot E'_j \approx M_{ij}$. If $M = EE'^T$, then for any orthogonal Q :

$$M = (EQ)(E'Q)^T \quad (13)$$

The factorization is determined only up to rotation. Community and degree information could be distributed across dimensions arbitrarily.

6.2 Barrier 2: Low-Rank Approximation

When embedding dimension $d < \text{rank}(M) \leq K + 2$, SGNS computes an approximate factorization. The rank-1 offset terms $\mathbf{r}\mathbf{1}^T + \mathbf{1}\mathbf{r}^T$ may or may not align with the top- d singular vectors of M . Their fate under projection is not analytically tractable.

6.3 Barrier 3: Stochastic Optimization

SGNS uses stochastic gradient descent, not SVD. The final embedding depends on:

- Random initialization
- Learning rate and schedule
- Order of training samples
- Number of negative samples

Different runs may converge to different local optima with different distributions of community/degree information across dimensions.

7 Summary and Implications

Pipeline Stage	Community Info	Separation Preserved?
Normalized adjacency S	In eigenvector direction	Yes (known)
Co-occurrence C	In similar eigenvectors	Yes (Prop. 1)
PMI matrix M	Block structure + offsets	No (additive)
SGNS embeddings E	Unknown	Cannot determine

The PMI transform is the critical juncture. Before it, community and degree are multiplicatively separable. After it, they are additively entangled.

7.1 Practical Implications

1. **When degree heterogeneity is high**, spectral methods with row-normalization have provable guarantees; Node2Vec does not.
2. **Normalizing Node2Vec embeddings** will not recover the clean separation that works for spectral methods—the underlying structure is different.
3. **Empirical questions remain:** Does the additive structure approximately preserve separation in practice? This requires experiments on DCSBM graphs with varying degree heterogeneity.

8 Conclusion

We traced community information through the Node2Vec pipeline on DCSBM graphs. While the co-occurrence matrix preserves the multiplicative eigenvector structure of spectral methods, the PMI transform converts this to additive structure. Combined with the non-uniqueness, approximation, and stochasticity of SGNS, this explains why Node2Vec resists the theoretical analysis possible for Laplacian Eigenmaps.

The insight is not that Node2Vec performs worse—empirically it often excels—but that its behavior on structured graphs cannot be predicted from theory alone. For applications requiring guarantees under degree heterogeneity, spectral methods remain the principled choice.

References

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